

1 Validation Check

1.1 Overview

The validation check of a dialog verifies whether the system can completely process all resulting events (in the current status). If the dialog cannot be processed, i.e. not all events have been checked successfully, the dialog is rejected without making changes to the system.

The validation checks evaluate dialog data, the generated events and the HYDRA data set and interpret the posting status. Some (partly optional) configurations are also integrated.

Examples of **validation checks for dialog data**:

- if an OP is logged on, the logged on OP must exist in the HYDRA data set,
- if an OP is logged on, the machine where the OP is logged on must exist in the HYDRA data set,
- if staff is logged on, the logged on person must exist in the HYDRA data set,
- when postings are performed for group workplaces, the postings must include the persons.

Examples of **validation checks for the posting status**:

- if a part quantity is uploaded for an OP, the OP must be logged on to the machine where the posting is performed,
- if staff is logged off, the logged off person must have been logged on to the machine where the posting is performed.

Examples of **validation checks for configurations**:

- if an OP is logged on, the person who performs the logon must be authorized to log on the OP,
- if a part quantity is uploaded for an OP, the overdelivery quantity may not be exceeded.

Some validation checks are explained in more detail in the sections that follow:

1.2 Validation checks regarding overdelivery or underdelivery

1.2.1 Overview

If part quantities are uploaded for an operation or if an operation is interrupted or logged off, a validation check can be performed for the quantities posted. It is checked if there is an overdelivery. When an operation is logged off, you can also check if there is an underdelivery.

By default, quantity checks in HYDRA are designed for the manual input of quantities. The quantity check is performed if a posting dialog includes a quantity $\neq 0$. The quantity check is also always performed if a quantity $\neq 0$ is posted. In case of an overdelivery, the check is also performed if the quantity has already been confirmed and a deviation reason has been entered.

The overdelivery/underdelivery check, also referred to as target quantity check, can be activated for operations or for persons.

1.2.2 Operation-related activation of the overdelivery/underdelivery check

You can activate a validation check for overdelivery or underdelivery for the operation. The following values/settings are relevant:

Underdelivery

Value in percent. The posted quantity can deviate from the target quantity by the percentage specified.

Example:

Target quantity of the operation: 120 items

Underdelivery: 84%

The actual quantity must not fall below 101 items.

Overdelivery

Value in percent. The posted quantity can deviate from the target quantity by the percentage specified.

Example:

Target quantity of the operation: 120 items

Overdelivery: 168%

The actual quantity must not exceed 201 items.

Reaction

If the limits specified in the fields overdelivery or underdelivery are exceeded, a warning or an error message can be issued. Possible values are:

"blank" no reaction

W warning

X error

If a warning is activated ("W"), you can confirm a quantity deviation on the Windows terminal by entering a deviation reason. If an error is activated ("X"), a quantity deviation is rejected.

Notes

By default, the operation-related validation check is performed for the actual yield quantity recorded (primary quantity unit). You can also configure the processing code and activate a validation check for scrap, rework or problem quantities. A validation check for alternative quantity units (e.g. secondary quantity, etc.) is not provided in the standard.

In general: DOS terminals generally reject all quantity deviations with an error, even if the option is set to "W".

The functionality, which is available for Windows terminals, does not exist on the MOC.

If you use the SAP interface "PP-PDC", limit values for overdelivery/underdelivery are transferred as absolute values. HYDRA converts these values to percentage values in the interface.

1.2.3 Person-related activation of the overdelivery/underdelivery check

In addition or as an alternative, you can also activate a person-related overdelivery/underdelivery check. For this purpose, the HR master provides the "target quantity check" option in the "BDE" tab. Possible values:

- 1) No check.
- 2) Order logoff: When an operation is logged off, the system checks if the current yield is between the specified minimum and maximum target quantity. Both quantities are also specified in the HR master for a person.
- 3) Underdelivery/overdelivery: All quantity postings are checked for overdelivery, when orders are logged off/partially uploaded and interrupted. When an operation is logged off, it is also checked for underdelivery.

You can confirm the validation check on the Windows terminal, if you enter a deviation reason.

Notes

The person-related validation checks are only performed for yield (primary quantity unit).

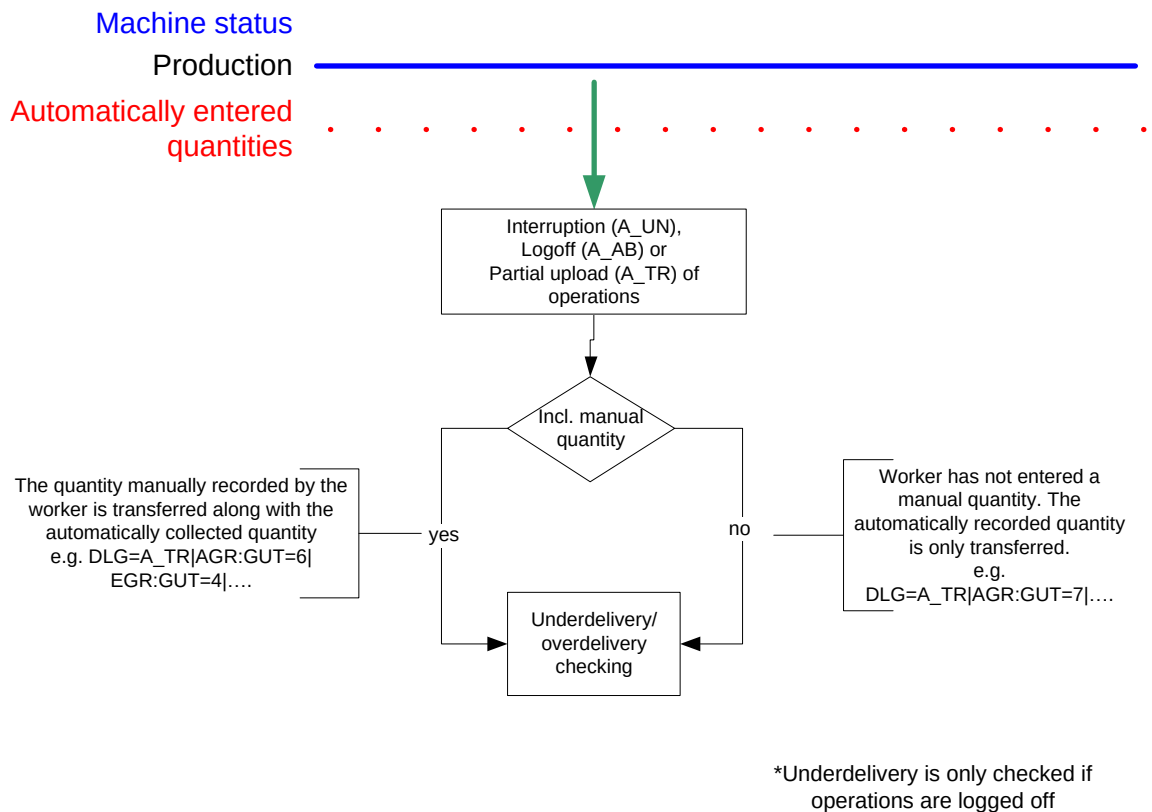
The limit values for overdelivery and underdelivery do not affect the person-related settings and are checked separately, if activated. If both validation checks are active, then first the person-related target quantity check is performed, then the operation-related check.

1.2.4 Overdelivery/underdelivery check with automatic recording of quantities

In HYDRA standard, the target quantity check is designed for the manual entry of quantities. It is only performed if a quantity $\neq 0$ is entered in the posting dialog. The paragraphs below describe different scenarios where the validation check is performed in connection with the automatic recording of quantities.

Scenario 1

If a quantity posting is performed for an operation in the active "production" status (interruption, logoff or partial confirmation/upload), the quantities that have been recorded automatically since the last posting (e.g. status change, operation posting, personnel posting) and the quantities that might have been recorded manually are sent to the server to be booked. A validation check is performed, because the command (DLG=...) sent to the server includes a quantity. It does not matter if the operator has manually entered a quantity $\neq 0$. The validation check is now performed for the quantity automatically recorded (AGR:GUT=...) and for the manually entered quantity (EGR:GUT=...=).



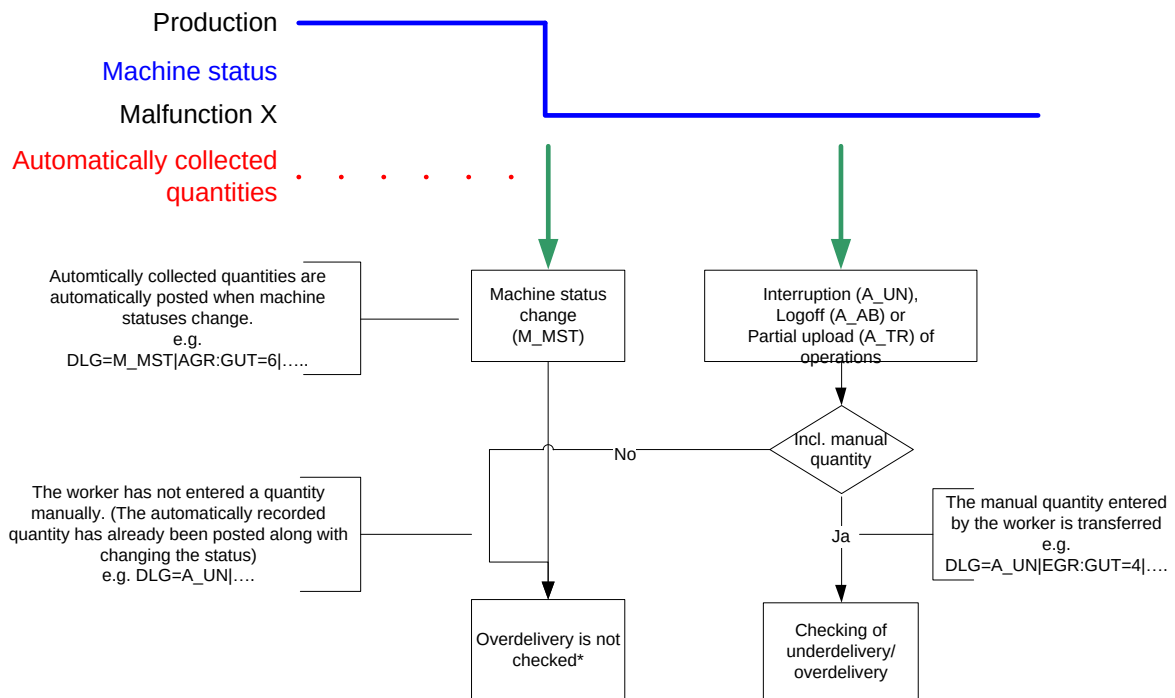
Scenario 2

If a machine switches from the production status to “malfunction”, the quantities recorded so far (AGR:GUT=...) are sent to the server with the status change (DLG=M_MST|...) to be booked. As this is an automatic posting (status change), a validation check is not performed.

If a posting is later on performed for the operation, a validation check is only performed if the operator manually enters a quantity $\neq 0$. This quantity is sent to the server with the posting (e.g. DLG=A_UN|EGR:GUT=...).

If the worker, for example, does not enter a quantity when the operation is interrupted (quantity = 0), no validation check is performed, because no quantity is sent to the server (DLG=A_UN|...).

As no automatic quantities have been recorded since the last status change, this posting does not include any automatic quantities, which could be checked.



Scenario 3

Automatic quantities can still be recorded, even if the machine is in the malfunction status and the production lock is set.

In this case, this automatic quantity is sent to the server along with the manual quantity entry (e.g. DLG=A_UN|AGR:GUT=...) and is checked for validity.

The validation check is performed as described in scenario 1. It does not matter if the operator has made manual quantity entries or not.

